Data Displays

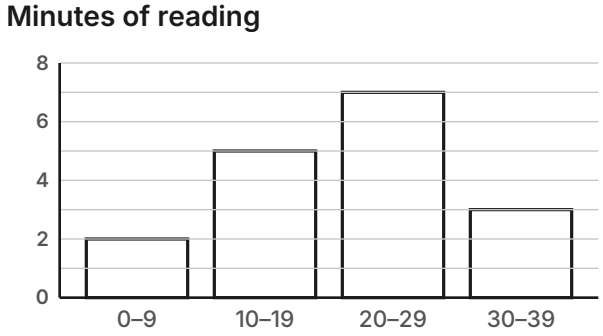
Ask a question with many answers. Then show the data.

A statistics question expects different answers

“How tall are the kids in my class?” Yes: many answers.

“How tall is Mia?” No: one answer.

A histogram: count the data in equal groups



Each bar is a group of numbers.

7 kids read 20 to 29 minutes.

The bars touch: the groups follow on.

One far-off number pulls the mean, not the median

2, 3, 3, 4, 28: mean 8, median 3

Your turn!

1, 4, 4, 6, 15: mean = 6 and median = 4

For teachers

Goal: statistics questions, X charts and histograms, and how a far-off value moves the mean (6.SP.1, 6.SP.4, 6.SP.5). Say “Does this question have one answer or many?”

Data Displays: Practice

Group the data. Show it. Read it.

Part 1 Statistics question? Many different answers? Circle yes or no.

1)	**How far can sixth graders jump?**	(yes) no
2)	**How many hours of sleep do kids in my class get?**	(yes) no
3)	**How tall is Mia?**	yes (no)
4)	**How tall are the trees in the park?**	(yes) no
5)	**What time does school start today?**	yes (no)
6)	**How many days are in a week?**	yes (no)
7)	**How many pets do families on my street have?**	(yes) no
8)	**How many legs does a spider have?**	yes (no)

Data Displays: Practice (continued)

Part 2 Count the groups How many numbers fall in each group?

9) **12, 25, 7, 18, 31, 22, 9, 15, 27, 3, 19, 24**

0–9 **3** 10–19 **4** 20–29 **4** 30–39 **1**

10) **45, 52, 61, 48, 57, 66, 50, 43, 59, 63**

40–49 **3** 50–59 **4** 60–69 **3**

11) **4, 11, 8, 15, 2, 13, 6, 9, 17, 12, 5, 10**

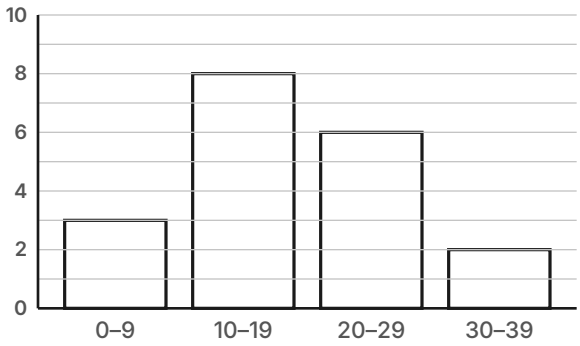
0–4 **2** 5–9 **4** 10–14 **4** 15–19 **2**

12) **33, 21, 38, 26, 29, 35, 24, 31, 37, 22**

20–24 **3** 25–29 **2** 30–34 **2** 35–39 **3**

Part 3 Read the histogram Each bar is a group.

13) **Minutes on the bus**



How many in 20–29? **6**

Most in which group? **10–19**

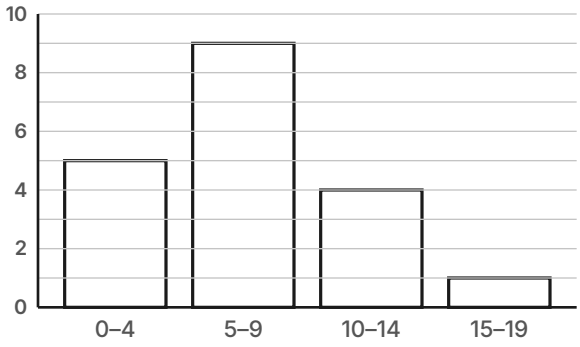
How many are 20 or more? **8**

How many in all? **19**

Data Displays: Practice (continued)

Part 3 Read the histogram (continued)

14) **Points in a game**



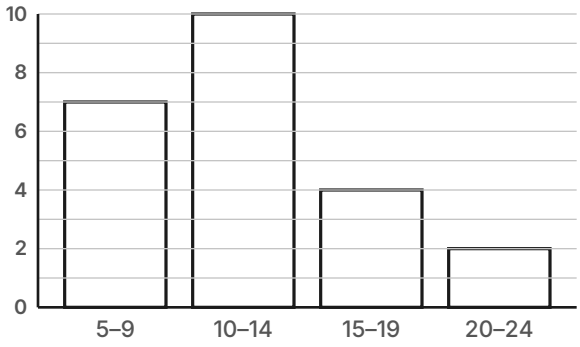
How many in 10–14? **4**

Most in which group? **5–9**

How many are 10 or more? **5**

How many in all? **19**

15) **Ages at the center**



How many in 15–19? **4**

Most in which group? **10–14**

How many are 15 or more? **6**

How many in all? **23**

Part 4 Make the X chart One X for each number.

16) **2, 0, 1, 3, 1, 1, 2, 0, 1**

Siblings

0	1	2	3
X	X	X	X
X	X	X	
	X		

mode = **1**

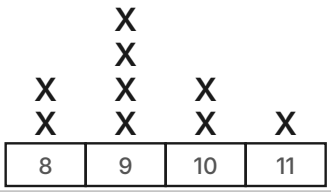
range = **3**

Data Displays: Practice (continued)

Part 4 Make the X chart (continued)

17) **9, 8, 10, 9, 9, 8, 11, 10, 9**

Hours of sleep

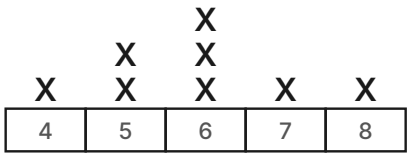


mode = 9

range = 3

18) **5, 6, 4, 6, 7, 5, 6, 8**

Shoe sizes



mode = 6

range = 4

Part 5 A far-off number Find both. Which one moved?

19) **3, 4, 4, 5, 24: mean =** 8 **and median =** 4

20) **10, 12, 12, 14, 52: mean =** 20 **and median =** 12

21) **1, 2, 2, 3, 17: mean =** 5 **and median =** 2

22) **6, 7, 8, 8, 31: mean =** 12 **and median =** 8

23) **20, 21, 21, 22, 66: mean =** 30 **and median =** 21

Data Displays: Practice (continued)

Part 6 More far-off numbers

24) 34, 35, 37, 36, 68: mean = 42 and median = 36

25) 14, 92, 13, 12, 14: mean = 29 and median = 14

26) 36, 34, 31, 31, 78: mean = 42 and median = 34

27) 22, 23, 19, 46, 20: mean = 26 and median = 22

28) 24, 29, 25, 97, 25: mean = 40 and median = 25

29) 21, 25, 95, 21, 23: mean = 37 and median = 23

30) 37, 55, 36, 37, 40: mean = 41 and median = 37

31) 9, 6, 8, 6, 76: mean = 21 and median = 8

32) 47, 10, 7, 5, 6: mean = 15 and median = 7

33) 8, 82, 11, 10, 9: mean = 24 and median = 10

Data Displays: Practice (continued)

Part 7 More means and medians

34) **mean of 41, 53, 20, 46 =** 40

35) **mean of 39, 15, 56, 26 =** 34

36) **mean of 25, 35, 50, 42 =** 38

37) **mean of 13, 28, 44, 39 =** 31

38) **mean of 34, 22, 60, 20 =** 34

39) **mean of 55, 23, 14, 40 =** 33

40) **mean of 56, 46, 45, 17 =** 41

41) **mean of 58, 47, 13, 50 =** 42

42) **mean of 22, 46, 59, 29 =** 39

43) **mean of 13, 33, 35, 15 =** 24

44) **mean of 32, 10, 17, 49 =** 27

45) **mean of 37, 44, 24, 27 =** 33

Data Displays: Practice (continued)

Part 8 Remember? Problems from earlier packets.

46) **mean of 8, 13, 9, 14 =** **11**

47) $6 \times 5 \times 2 + 2 \times 4 =$ **68**

48) $3\frac{1}{2} \times 2 \times 4 =$ **28**

49) $x + 47 = 92$: $x =$ **45**

50) $3{,}456 \div 12 =$ **288**

51) -15 **<** 2

Part 9 Word problems Which measure? Then find it.

52) Test scores: 82, 90, 75, 90, 88. What is the median?

Number sentence **75, 82, 88, 90, 90**

Answer **88** points

Draw or show your work

53) Kids read 3, 5, 4, 5, 8 books. What is the mean?

Number sentence **3 + 5 + 4 + 5 + 8 = 25, 25 ÷ 5 = 5**

Answer **5** books

Draw or show your work

54) In a histogram the bars show 4, 9, 6 and 1 kids. How many kids in all?

Number sentence **4 + 9 + 6 + 1 = 20**

Answer **20** kids

Draw or show your work

Data Displays: Practice (continued)

Part 9 Word problems (continued)

55) Heights: 50, 52, 52, 55, 56 in. What is the range?

Number sentence **56 − 50 = 6**

Answer **6** in

Draw or show your work

56) Challenge: Wait times 2, 3, 3, 4 and 38 minutes. Mean minus median?

Number sentence **10 − 3 = 7**

Answer **7** minutes

Draw or show your work

Check-In: Data Displays

Show what you know. No help this time.

Mastery: 4 of 5 (80%)

Part 1 Statistics question? Circle yes or no.

1) **How much rain fell each day in May?**

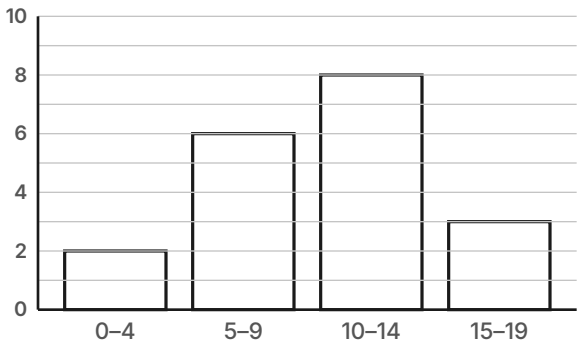
(yes) no

2) **How old is my teacher?**

yes (no)

Part 2 Read the histogram

3) **Laps run**



How many in 15–19? **3**

Most in which group? **10–14**

How many are 10 or more? **11**

How many in all? **19**

Part 3 Count the groups

4) **14, 27, 8, 21, 33, 16, 29, 5, 24, 18, 36, 11**

0–9 **2** 10–19 **4** 20–29 **4** 30–39 **2**

Part 4 A far-off number

5) **5, 6, 6, 7, 31: mean = 11 and median = 6**